Mauro Bagnoli

Backend & Platform Engineer — Python · AWS · Terraform

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Summary

Backend and infrastructure engineer with 9+ years in software, focused on serverless, event-driven systems on AWS. Currently backend and infrastructure owner of a multi-market agronomic platform at Bayer: split a Python monolith into 17 independently releasable components, designed the event bus that connects them, and drove the cross-team architecture that lets other teams reuse them. Comfortable owning a system end to end — domain modelling, Terraform, CI/CD, observability, and the developer tooling around it.

Experience

Senior Backend & Infrastructure Engineer at Kopius (contracted to Bayer)

2024 — Present · Remote (Spain)

Backend and infrastructure owner of Prediview, an agronomic platform that turns weather and phenology models into plot-level disease risk and treatment recommendations for 602 growers across 2,527 fields (~17,500 ha) in France and Germany.

- Split a Python monolith into **17 independently releasable components**, each colocating application code and its Terraform infrastructure; the API alone has shipped **91 independent releases** since, with emails, migrations and infrastructure no longer moving in lockstep. Migrated the build from Poetry to uv/hatchling to keep infra and code in a single component folder.
- Designed and implemented an event-driven message bus on **EventBridge + SQS** with DLQs, retry policies and Teams alerting, plus a shared installable Pydantic contract package consumed by every component. Nine months in production across eight queues have produced **a single DLQ incident**; when an external provider API went down for seven days, queues drained cleanly with no data loss.
- Raised backend test coverage above **85%** and enforced it as a CI gate, backed by **1,611 tests across 497 files**; refactored logging so error level means developer action, and closed security gaps in token-expiry verification and committed secrets inherited from before handover. User-filed bug reports fell from a peak of 17 in April to 1 in July.
- Led the first **Germany** season end to end — 20 distributors, 136 growers, 463 fields and ~4,100 ha between February and June 2026 — integrating an in-house phenology model with external providers at a **99.96% field-registration success rate** (1 failure in 2,528).
- Independently captured and analysed a full season of external model-provider responses, visualising **1,392 treatment windows** and their durations, overlaps and risk profiles for the product team; the analysis exposed providers returning high risk on window-transition days, producing unexpected treatment recommendations, and gave PMs the evidence they needed to take the issue to the providers directly.
- Pushed for and delivered a backoffice tool that ended developer-run data entry: engineers had hand-written and executed **25 distributor onboarding scripts** between January and March 2026, roughly one every three days, a workflow now owned end to end by the country-representative PMs. Onboarded the **UK** market with its own roles and permissions so the local team could prepare its campaign unaided.
- Proposed and drove a **cross-team component-sharing model** (versioned service repo + template repo) letting other teams deploy isolated copies without sharing environment variables or cost attribution; migrated the SMS/push messaging service from TypeScript to Python as the first shared component.
- Set the team's AI-assisted development conventions, authoring the repository's Copilot guidelines file so generated code follows project architecture and testing standards; built a Copilot-backed DLQ analyzer, a hot-reload Jinja2 email-template preview server, and a fake model provider for off-season development and user training.

Full Stack Developer at White Prompt (staff augmentation)

Jun 2022 — 2024

Client: **Invicara** (Jun 2022 — Mar 2024), a Building Information Modeling operations platform, alongside ~20 engineers, designers and QA.

- Migrated the codebase from untyped JavaScript to **TypeScript**, moving a class of integration errors from runtime to compile time across a platform maintained by twenty people.
- Built the integration between the web application and the company's microservices architecture with Node.js.
- Led a large refactor of the existing codebase, introducing facade, singleton and builder patterns and the first comprehensive test suites the application had.
- Modernised the build and developer workflow — upgraded the module bundler, and introduced linting and shared code standards — then owned onboarding and code gatekeeping so they held as the team grew.

Full Stack Developer at Magoya Software

Jun 2020 — 2022

- Delivered features end to end on a Django/DRF and React/Next.js stack in a team of ten, taking on progressively broader ownership across the two years.
- Extended the API layer beyond REST with Graphene (GraphQL), and typed the front end with TypeScript.
- Containerised the development environment with Docker and Docker Compose, wired up Sentry for error tracking, and deployed to Google Cloud.

Full Stack Developer at Infoxel

Jan 2020 — Jun 2020

- Django REST Framework, ElasticSearch, TypeScript and React in a team of six.

Software Developer & Data Assistant at Fundación Conocimiento Abierto

Jan 2018 — Dec 2019

- Data pipelines and visualisation with Python, Scrapy, Pandas, D3 and React, working daily with multidisciplinary teams.

Web Developer at Editorial Nodo Gráfico

Dec 2016 — Jan 2018

Skills

Languages — Python, TypeScript/JavaScript, SQL, Bash

Cloud & Infrastructure — AWS (Lambda, API Gateway, EventBridge, SQS, DynamoDB, RDS, VPC, IAM, CloudWatch, X-Ray), Terraform, Docker, Ansible, Nginx, GCP

Backend — Serverless and event-driven architecture, domain-driven design, SQLAlchemy, Pydantic, FastAPI, Django/DRF, REST, GraphQL, message buses, unit-of-work and transactional messaging

Practice — CI/CD (GitHub Actions, Azure DevOps), testing and coverage gates, observability and alerting, security best practices, AI-assisted development (GitHub Copilot, repo-level agent guidelines), technical mentoring and onboarding

Frontend — React, React Native, Next.js, Vue, TanStack Query, Tailwind

Education

University of Buenos Aires

2013 — 2020 · Degree in Communication Sciences

Digital House

2018 — 2019 · Full Stack Development and Data Science

Aves Argentinas

2017 — 2019 · Naturalist Interpreter

Projects

Birds of Mallorca — React Native app for identifying Mallorcan birds by photo, with multilingual support for residents and visitors. Live on the App Store and in active development.

Natujuegos — Open-source quiz game for learning Argentine bird species (Vue, Express).

Interests

Birdwatching, trekking, scuba diving, travelling and reading.